

The Global Internet Gap

Access, Gender & Growth (2000–2022)

*A World Bank World Development Indicators (WDI) Analysis
Capstone Project · Week 8 · AnalystLab Africa Data Analytics Internship*

1. Objective / Problem Statement

This capstone project investigates global disparities in technology and internet adoption between 2000 and 2022, using the World Bank's World Development Indicators (WDI) dataset. The goal is to quantify how internet access, mobile connectivity, and broadband adoption have evolved worldwide, identify where meaningful gaps remain — across regions, income levels, and gender — and translate these patterns into actionable insights and recommendations for policymakers, development organizations, and technology investors.

The analysis addresses three guiding questions:

- How has global internet, mobile, and broadband adoption changed over the past two decades?
- Which regions and income groups lead or lag in technology access, and how large is the gap?
- Is there a meaningful gender gap in internet usage, and does it vary by a country's income level?

2. Dataset Description

The dataset is sourced from the World Bank's World Development Indicators (WDI) — a comprehensive global database covering economic, social, and environmental indicators for over 200 countries. The full raw file contains approximately 1,500 distinct indicators across all countries from 1960 to 2025.

For this project, six indicators related to the Technology & Internet Adoption theme were selected:

- Individuals using the Internet (% of population)
- Individuals using the Internet, female (% of female population)
- Individuals using the Internet, male (% of male population)
- Mobile cellular subscriptions (per 100 people)
- Fixed broadband subscriptions (per 100 people)
- Fixed telephone subscriptions (per 100 people)

The World Bank's country metadata file (WDICountry.csv) was used to distinguish genuine countries from regional and income-group aggregates (e.g., “World,” “Arab World,” “High income”), and to attach a Region and Income Group classification to each country for segmentation.

3. Data Cleaning Process

The raw WDI file (189MB, ~1,500 indicators, all countries and aggregates, 1960–2025) was processed through the following steps:

1. Filtered the full dataset down to the 6 indicators relevant to the Technology & Internet Adoption theme, reducing the file to 1,590 rows.
2. Reshaped the data from wide format (years spread across 66 columns) to long format (one row per country-indicator-year combination) using pandas' melt() function, and dropped rows with no reported value, producing 42,880 rows.
3. Merged the dataset with the WDICountry metadata file to retain only genuine countries (excluding regional/income aggregates) and to attach Region and Income Group columns, narrowing the data to 35,667 rows across 214 countries.

- Filtered the year range to 2000–2022, excluding pre-2000 data (largely zero or missing, since internet and mobile technology were not yet widespread) and 2023–2025 (incomplete reporting), resulting in a final analysis-ready dataset of 20,404 rows.

The table below summarizes this progression:

Stage	Description	Row Count
Raw file	All indicators, all countries, 1960–2025	~6M+ rows (full file)
Indicator filter	6 technology indicators only	1,590
Reshape to long format	Years unpivoted into rows; blanks dropped	42,880
Country filter	Aggregates removed; Region/Income Group added	35,667
Year filter	Restricted to 2000–2022	20,404 (final)

4. Analysis Methodology

Exploratory analysis was first conducted in Python (pandas) to establish key trends and validate findings before building the interactive dashboard. This included calculating global year-over-year averages, grouping by Region and Income Group, and comparing gender-specific indicators. These findings were then used to design a Power BI dashboard consisting of four KPI cards, five visualizations (a global trend line, a regional comparison bar chart, a mobile-vs-broadband trend comparison, a gender gap by income group chart, and a filled world map), and three interactive slicers (Year, Region, Income Group) to allow further exploration.

5. Key Findings

Metric	Value
Global Internet Usage, 2000	8.73%
Global Internet Usage, 2022	70.24%
Highest-Access Region (2022)	North America — 93.36%
Lowest-Access Region (2022)	Sub-Saharan Africa — 38.15%
Regional Gap (2022)	55.2 percentage points
Global Gender Gap, 2022 (male – female)	1.32 percentage points
Gender Gap, Lower-Middle Income Countries (2022)	7.57 percentage points
Gender Gap, High-Income Countries (2022)	0.76 percentage points
Avg. Mobile Subscriptions (2022)	111.11 per 100 people
Avg. Fixed Broadband Subscriptions (2022)	17.40 per 100 people

6. Insights

- Global internet access grew more than eight-fold between 2000 and 2022 (8.73% to 70.24%), with growth accelerating notably after 2015 — likely reflecting the fall in smartphone and mobile data costs during that period.
- A substantial regional digital divide persists: North America's internet access rate (93.36%) is roughly 2.4 times that of Sub-Saharan Africa (38.15%), a 55-point gap that has direct implications for education, commerce, and access to information.
- Mobile technology has vastly outpaced fixed broadband adoption globally (111.11 vs. 17.40 subscriptions per 100 people), indicating that mobile networks — not fixed-line infrastructure — are the primary way most of the world connects to digital services.
- While the global gender gap in internet usage appears small (1.32 points), this figure masks significant inequality: Lower-middle income countries show a gender gap nearly 10 times larger (7.57 points) than High-income countries (0.76 points), revealing that gender-based digital exclusion is concentrated in specific economic contexts rather than being a uniformly small, worldwide issue.

5. Low-income countries were absent from gender-disaggregated internet data for 2022, a data availability gap that itself signals under-investment in monitoring digital inclusion in the world's poorest economies.

7. Recommendations

1. Prioritize mobile-first infrastructure investment in low-access regions, particularly Sub-Saharan Africa, since mobile networks have proven to be the dominant and most scalable channel for closing access gaps, rather than costly fixed-line broadband expansion.
2. Target digital inclusion programs specifically at lower-middle income countries, where the gender gap in internet access is largest — blanket global gender-parity initiatives risk misallocating resources away from where the gap is most severe.
3. Improve data collection on gender-disaggregated technology metrics in low-income countries, so that future policy decisions in these regions can be evidence-based rather than made in the absence of reliable data.
4. Use regional benchmarking (e.g., comparing a country against its regional peers rather than only the global average) when setting national digital inclusion targets, since the data shows regional context matters more than a single global benchmark suggests.
5. Continue monitoring the mobile-to-broadband transition, as regions currently reliant on mobile-only access may face limitations for data-intensive applications (e.g., remote work, online education) that benefit from fixed broadband infrastructure.

8. Conclusion

This analysis of two decades of World Bank data reveals a story of remarkable global progress in technology adoption, tempered by persistent and unevenly distributed gaps. While the world as a whole has become dramatically more connected since 2000, that progress has not been shared equally — across regions, income levels, or gender. Mobile technology has been the primary engine of this growth, reaching populations that fixed infrastructure has not. The clearest opportunity going forward lies in targeted, mobile-first investment in the lowest-access regions, combined with better data collection to illuminate — and eventually close — the gender and income-based gaps that remain hidden within global averages.